NORMA INÉS LLERGO SÁNCHEZ

Software Engineering Student | Programming, Data Science, Machine Learning & IoT Puebla, Mexico | normaines.llergo@gmail.com | Portfolio | GitHub | LinkedIn

PROFILE

Software Engineering student focused on programming, data analysis, applied machine learning, computer vision and IoT. Experience includes Java desktop applications, Python model development, C/C++ application logic, web prototypes and sensor based monitoring systems.

EDUCATION

Universidad Popular Autónoma del Estado de Puebla (UPAEP) **Aug 2024 - Expected Jul 2028** *B.S. in Software Engineering | Puebla, Mexico*

EXPERIENCE

Texas Ceta Canyon Camp & Retreat Center **Summer 2026** *Seasonal Retail Operations | Canyon, Texas*

• Managed inventory records and operational reporting for the camp gift shop and concession stand. **Research Assistant - Applied Technology Projects** **Aug 2024 - Present** • Contributed to university research-oriented projects involving food-fermentation monitoring and river-pollution analysis, including collaboration with biology faculty.

VOLUNTEER EXPERIENCE

IBM Student Ambassador - Student Ambassador Program with IBM SkillsBuild Sep 2026 - Jan 2027 *Volunteer | IBM*

• Volunteer member of the IBM SkillsBuild Student Ambassador Program, a student community program focused on technology and career-readiness learning.

• Completed IBM SkillsBuild certifications in building an AI project and in open source (see Training and Certificates).

SELECTED TECHNICAL PROJECTS

IoT Fermentation Monitoring Prototype Repository

Raspberry Pi 4 | Python | Ubuntu Linux | GPIO/SPI | DHT11 and MQ sensors | JSON/HTTP

• Designed and implemented the hardware and software for monitoring temperature, humidity, alcohol and gases, displaying readings on an LCD and transmitting timestamped data through an Azure IoT pipeline. Reported 98% agreement with documented reference data from other fermentation systems; prepared the project for ExpoCiencias Puebla 2025. **River Pollution Object Detection** Repository

YOLOv8 | Stable Diffusion | Google Colab | ONNX Runtime Web

• Independently generated more than 4,000 synthetic images of bottles, water, trash and plants in Google Colab to train a YOLOv8 model. Reported 82% mAP; released a browser prototype that identifies bottle, trash, water and plant classes with visual overlays and JSON output.

Diabetes Risk Prediction - Team project Repository

Python | pandas | scikit-learn | XGBoost | 100,000 records

• Led data cleaning, exploratory data analysis and training of KNN, Random Forest and XGBoost models; reported

approximately 97% predictive accuracy, with the repository notebook recording 97.22% mean accuracy for XGBoost under stratified 5-fold cross-validation.

Exoplanets Lab - NASA Space Apps Challenge team project Repository **Oct 2025** *Python data analysis | Astro | TypeScript | HTML Canvas*

- Performed exploratory analysis of an exoplanet dataset and supported the design of an interactive classifier/simulator; the team delivered an educational web application with a transit-method simulator, quiz and exploration section.

ADDITIONAL PROGRAMMING PROJECTS

Library Management System Repository

Java | Swing | OOP | inheritance | collections | exceptions

- Developed both the Java interface and system logic for catalog search, users, loans, returns, reservations, fines, physical condition history and statistics, using interconnected classes and inheritance.

Autonomous Line-Following Robot Repository **Apr 2024** *C# | infrared sensors | embedded navigation logic*

- Programmed navigation logic and assembled the hardware for an autonomous infrared-sensor line follower for the Ingeniobot competition.

C++ Application Logic Repository

C++ | data structures | functions | records | geometric transformations

- Implemented back-end logic for academic projects including Hangman, figure and space transformations, and a hotel registration system; visual interfaces were created with assistance from code-generation tools.

COMPETITIONS AND COMMUNITY LEADERSHIP

- Silver Medal, 18th place among 600 participants - 3rd Mexican Girls' Olympiad in Informatics (OFMI), Nov 2023.
- Honorable Mention, 53rd place among 161 participants - 2nd OFMI, Feb 2023.
- Organizing committee member for the 4th OFMI national competition (2024); teaches competitive programming in C++ .
- Active participant in the International Collegiate Programming Contest (ICPC) community since May 2025.
- IBM SkillsBuild: Build an AI project that makes your resume stand out (3Q 2026) ; What is open source?

TRAINING AND CERTIFICATES

- Facilitator - Ubuntu Leaders Academy, Instituto Padre António Vieira / UPAEP, five-day program, Jan 2024. • Liderazgo SIC - Avanzado (CENEVAL, Apr 2026); Personal Branding, 6 h (UDEM, Dec 2025); Introduction to Intelligent Bionic Androids, 120 min (P4H Bionics, Mar 2024).
- Food Management and Child Safety training - Ceta Canyon, 2026; TOEFL iBT preparation course certificate - Magoosh.

TECHNICAL SKILLS

Declared proficiency: Proficient in C/C++, Git and MySQL; intermediate Java, Python, PL/SQL and Figma; familiar with JavaScript, C# and HTML/CSS.

Data and AI: pandas, NumPy, scikit-learn, XGBoost, YOLOv8, ONNX Runtime, EDA, classification metrics, data augmentation and Google Colab.

Platforms: Git/GitHub, MySQL, FastAPI, Next.js, Astro, Raspberry Pi, Azure IoT, Linux, GPIO/SPI and JSON/HTTP.

Languages: Spanish (native); English (C1; Magoosh TOEFL iBT preparation); French(self reported A2).

Interests: Science-fiction reading, swimming, OSU, pastry baking and kawaii character illustration.